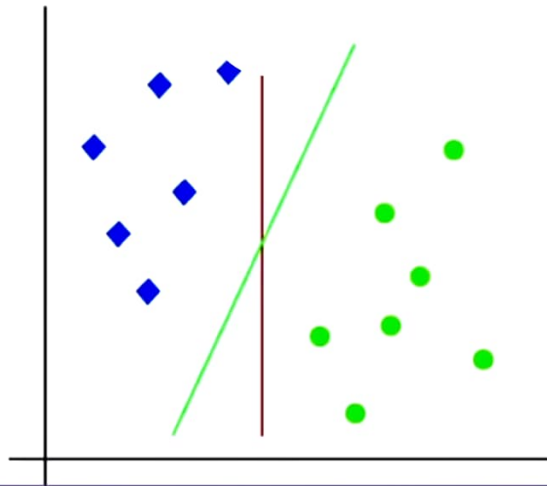
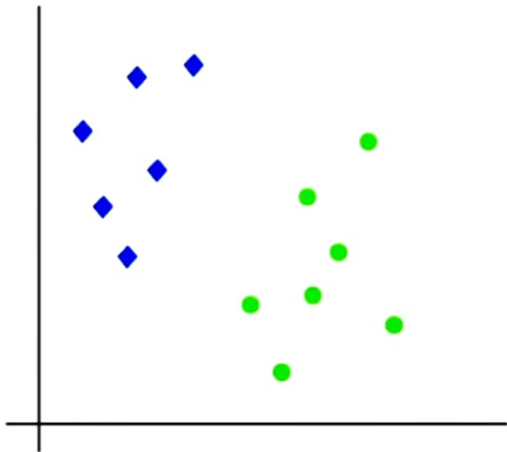


Support Vector Machine – Introduction

- Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems.
- However, primarily, it is used for Classification problems in Machine Learning.
- The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future.
- This best decision boundary is called a hyperplane.

Support Vector Machine – Hyperplane

- There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but we need to find out the best decision boundary that helps to classify the data points.
- This best boundary is known as the hyperplane of SVM.



Support Vector Machine – Hyperplane

- The dimensions of the hyperplane depend on the features present in the dataset, which means if there are 2 features (as shown in image), then hyperplane will be a straight line.
- And if there are 3 features, then hyperplane will be a 2-dimension plane.
- We always create a hyperplane that has a maximum margin, which means the maximum distance between the data points.

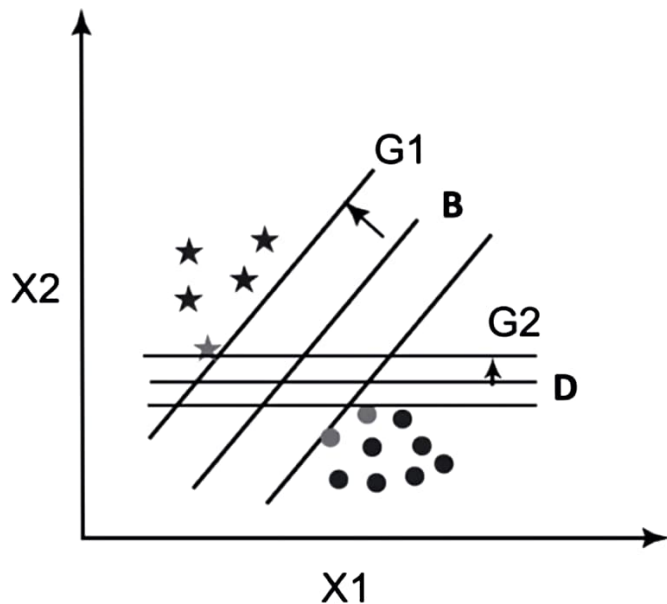
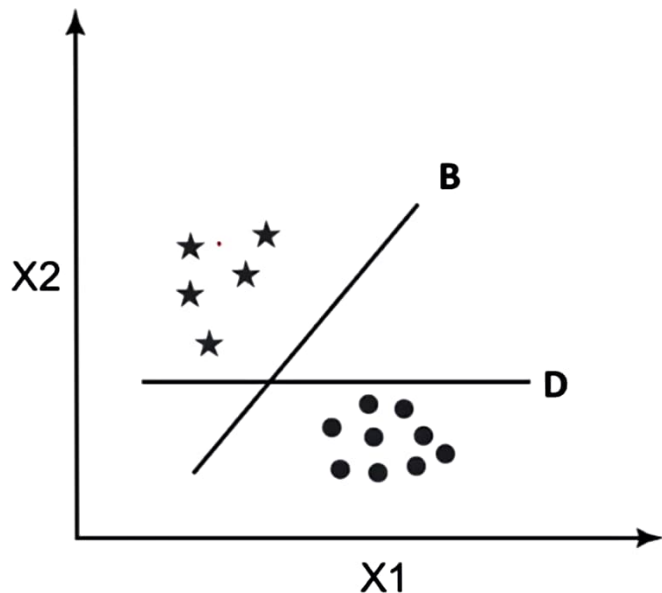
Support Vector Machine – Types of SVM

SVM can be of two types:

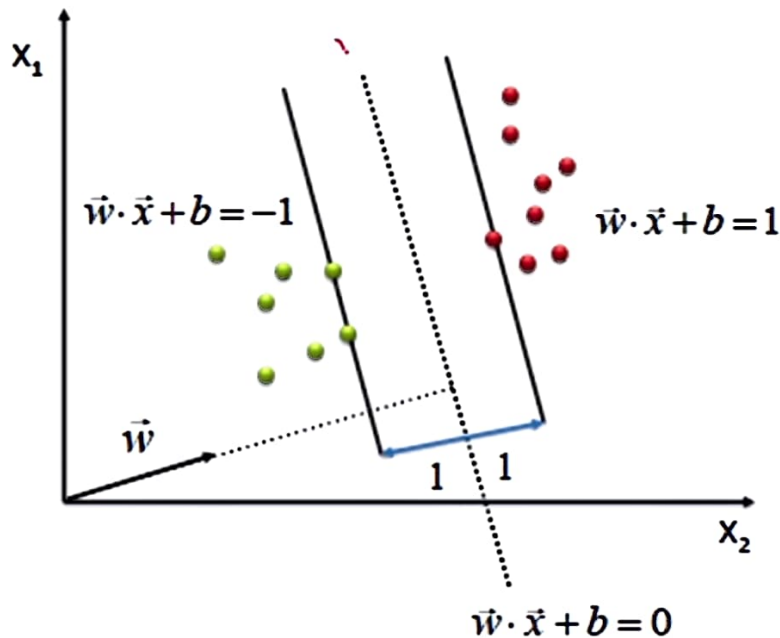
1. **Linear SVM:** Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.
2. **Non-linear SVM:** Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Support Vector Machine – Types of SVM

- Linear SVM:



Support Vector Machine – Types of SVM



$$\max \frac{2}{\|\vec{w}\|}$$

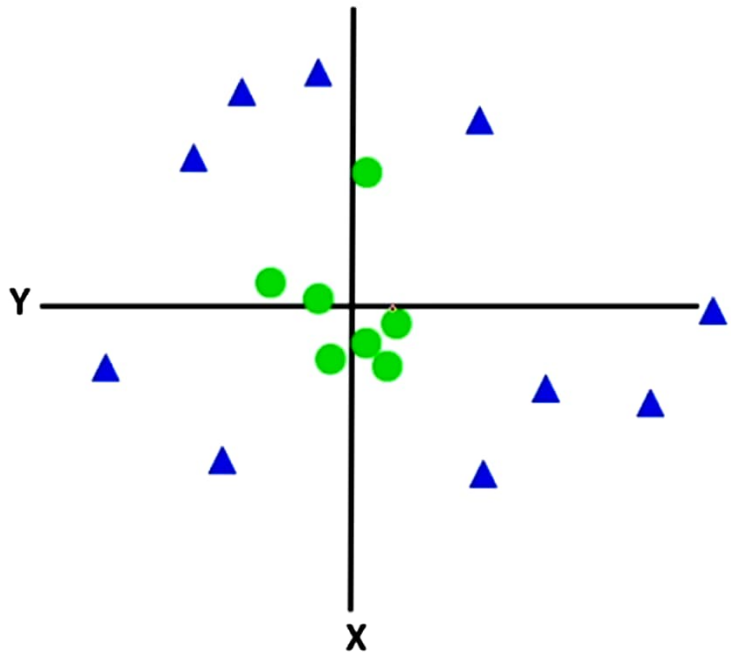
s.t.

$$(\vec{w} \cdot \vec{x} + b) \geq 1, \forall \vec{x} \text{ of class 1}$$

$$(\vec{w} \cdot \vec{x} + b) \leq -1, \forall \vec{x} \text{ of class 2}$$

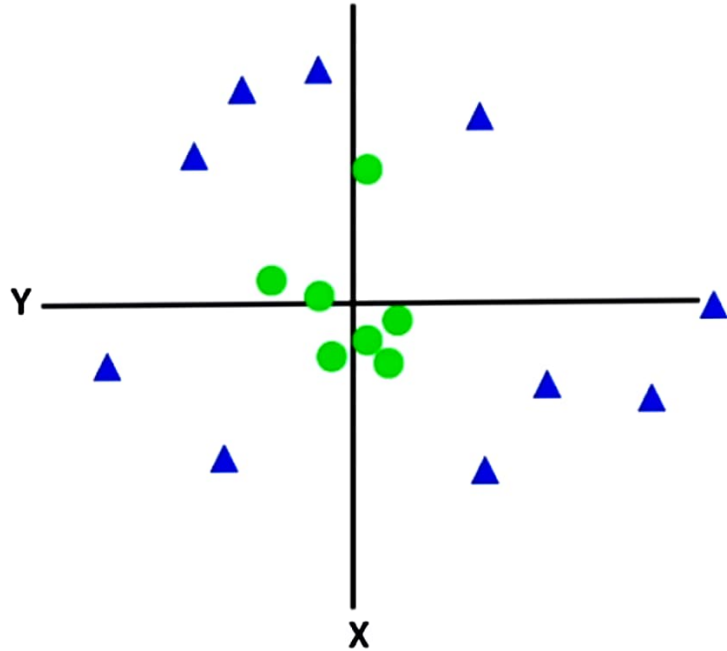
Support Vector Machine – Types of SVM

- **Non-Linear SVM:**

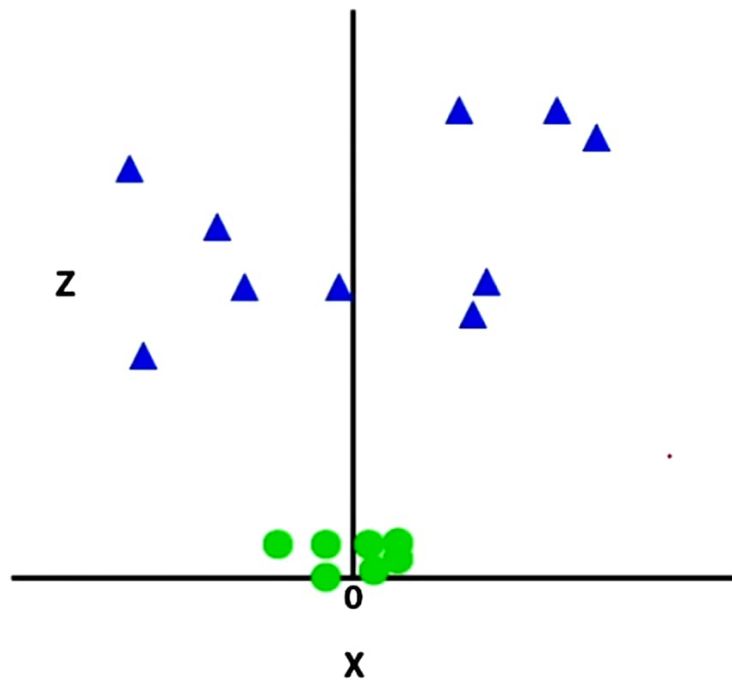


Support Vector Machine – Types of SVM

- **Non-Linear SVM:**
- If data is linearly arranged, then we can separate it by using a straight line, but for non-linear data, we cannot draw a single straight line.
- So to separate these data points, we need to add one more dimension.
- For linear data, we have used two dimensions x and y , so for non-linear data, we will add a third dimension z .
- It can be calculated as: $\mathbf{z} = \mathbf{x}^2 + \mathbf{y}^2$



Support Vector Machine – Types of SVM



Support Vector Machine – Types of SVM

